Fruit Twin GPT — Dashboard Operations & Workflow Guide

A Digital Twin-Driven Multimodal Framework for Early Fruit Quality & Yield Prediction

1. Executive Overview & System Architecture

Fruit Twin GPT is a production-grade full-stack machine learning system combining **Vision-Language Models (VLMs)**, **agronomic sensor telemetry**, and **dynamic Digital Twins** to predict fruit quality (°Brix sweetness, firmness, shelf life) and total tree harvest load (fruit count and yield weight) weeks before commercial harvest. The platform eliminates destructive sampling and manual repetitive paperwork through autonomous sensory tracking and 1-click multimodal ML inference.

Component	Technology Stack	Core Responsibilities
Client Dashboard	React.js, Vite, Recharts, Lucide	Responsive web interface, 5s workflow carousel, real-time telemetry rendering, 1-click model execution.
Backend Gateway	Java 21, Spring Boot 3.5, Security, JPA	JWT authentication, Digital Twin state lifecycle, REST controllers, observational persistence.
ML Inference Engine	Python 3.12, FastAPI, Scikit-learn, PIL	Multimodal fusion (ViT chroma features + agronomic GBM), multi-target regression & grading.
Relational Persistence	MySQL 8.0, Flyway Migrations	ACID storage of plant trees, time-stamped observations, prediction history, and harvest evaluations.

2. Comprehensive Guide to Dashboard Left Sidebar Options

The left sidebar provides intuitive, role-based navigation across the entire Digital Twin lifecycle. Below is the in-depth specification for each option:

Option 1: Overview (Mission Control & Telemetry Summary)

- **What it is:** The centralized executive dashboard aggregating tree health, quality readiness, and quick-action prediction controls across all orchard blocks.
- **What is the use:** Gives growers and agricultural researchers an instantaneous health check of orchard blocks without having to drill into individual plant records.
- **What it shows:** Active tree count, average fruit Brix index, dynamic 3-stage 5-second workflow carousel, real-time readiness gauge (88/100 score), and immediate 'Predict Fruit Quality & Yield' action.
- **How it works:** Queries /api/plants and /api/predictions, aggregates rolling averages, and powers real-time reactive updates.

Option 2: Plant Twins (Tree Registry & Asset Inventory)

- **What it is:** The digital asset registry where biological fruit trees are given permanent, unique virtual representations (Digital Twins).

- **What is the use:** Enrolls individual trees with botanical metadata (Plant Code, Crop, Variety, Planting Date, Location/Block) and baseline measurements.
- **What it shows:** Interactive list of registered trees, crop cultivars, planting calendar, sector/block identifiers, and individual tree prediction shortcuts.
- **How it works:** Submits plant profiles via POST /api/plants. Automatically initializes the tree's Digital Twin in MySQL and provisions its baseline sensory state.

Option 3: Observations (Automated Telemetry Logs)

- **What it is:** The sensory telemetry repository capturing environmental and morphological parameters across the fruit development cycle.
- **What is the use:** Eliminates manual, repetitive paperwork. Observations are **automatically maintained** whenever a tree is registered or when predictions are computed.
- **What it shows:** Chronological sensor logs: Soil Moisture (%), Soil pH, Canopy Temperature (°C), Ambient Humidity (%), Fruit Weight (g), Fruit Diameter (cm), and linked vision photography.
- **How it works:** Telemetry is synchronized with the tree record via POST /api/plants/{id}/predict and stored in the observations table linked by foreign key.

Option 4: ML Predictions (Multimodal Model Inference Engine)

- **What it is:** The advanced artificial intelligence hub executing trained multimodal fusion pipelines combining fruit imagery and agronomic features.
- **What is the use:** Provides non-destructive, explainable pre-harvest quality grading and tree fruit load forecasting weeks ahead of harvest.
- **What it shows:**
- **Quality Index (°Bx):** Predicted soluble solids content (e.g. 15.4 °Bx) with ±0.35 uncertainty bound.
- **Commercial Grade:** Export Grade A, Premium Grade B, or Commercial Processing.
- **Confidence Score:** Model certainty percentage (e.g. 94.6%).
- **Estimated Tree Fruit Load:** Total fruit yield forecast (e.g. 175–195 fruits, ~38 kg/tree).
- **Fruit Weight & Firmness:** Predicted harvest weight (215 g) and firmness index (34 N).
- **Projected Shelf Life:** Ambient storage viability (16 days).
- **Target Harvest Date:** Projected biological maturity date (e.g. 2026-11-02).
- **VLM Plain-English Explanation:** Agronomic rationale generated by the Vision-Language Model.
- **How it works:** Triggered by 'Predict Fruit Quality & Yield'. The Spring Boot API forwards normalized tree telemetry and image paths to the FastAPI inference engine (POST /predict/multimodal), evaluates the late-fusion GradientBoosting model ($R^2=0.941$), commits the prediction, and updates the Twin timeline.

Option 5: Twin History (Digital Twin Lifecycle & Temporal Trajectories)

- **What it is:** The temporal memory of the tree tracking its biological trajectory from early fruit set to harvest.
- **Why it is needed:** *Fruit development is a cumulative physiological process.* Sugar accumulation and fruit enlargement cannot be accurately assessed from a single isolated snapshot. Historical tracking reveals whether Brix is accelerating, plateauing, or suffering from moisture/heat stress.
- **What it shows:** Unified chronological timeline intertwining every physical sensor capture, fruit photo, and linked ML prediction event with model versioning stamps.
- **How it works:** Queries /api/plants/{id}/observations and /api/plants/{id}/predictions, merges records chronologically, and visualizes longitudinal biometric curves.

Option 6: Harvest & Evaluation (Ground Truth Benchmarking)

- **What it is:** The model validation and error analysis laboratory closing the loop between AI predictions and physical outcomes.
- **What is the use:** When fruit is picked, growers record the ground-truth refractometer Brix and scale weight to measure ML model accuracy.
- **What it shows:** Actual vs. Predicted comparison table, Mean Absolute Error (MAE = 0.32 °Bx), Percentage Error (e.g. 1.3% deviation), and residual error graphs.
- **How it works:** Dispatches POST /api/harvests, computes $| ext{Actual} - ext{Predicted}| / ext{Actual} \times 100\%$, and logs verification benchmarks.

3. Why 'Digital Twin History' is Essential in ML Precision Agriculture

A **Digital Twin** in horticulture is more than a 3D model or static database row—it is an evolving, bidirectional computational counterpart of a physical organism. Maintaining Twin History is critical for three scientific reasons:

1. Physiological Accumulation	Sugar synthesis (°Brix) depends on cumulative solar radiation and heat units (Growing Degree Days). Single snapshots miss whether photosynthesis is optimal or restricted.
2. Anomaly & Stress Detection	Comparing current sensor telemetry against historical baselines immediately detects microclimate anomalies (e.g. root-zone dryout, vapor pressure deficit spikes) before visible fruit damage occurs.
3. Model Calibration & Continuous Learning	By pairing historical sensor sequences with final measured harvest quality, the multimodal fusion models can be continuously fine-tuned and adapted across seasons and microclimates.

4. End-to-End Operational Workflow Playbook

Step 1 — Secure Authentication: User signs in via email/password. System validates credentials and issues a signed JWT token authorizing dashboard operations.

Step 2 — Register Fruit Plant: User opens 'Register Plant', provides Plant ID (e.g. MANGO-003), crop cultivar, block location, baseline fruit measurements, and fruit photo. The system creates the plant record and initializes the Digital Twin.

Step 3 — Automated Telemetry Logging: Field sensors and baseline photos are automatically captured and linked to the Digital Twin. No manual observation entry is required.

Step 4 — 1-Click Multimodal ML Prediction: User clicks 'Predict Fruit Quality & Yield'. The FastAPI ML service combines visual features and agronomic measurements to predict Brix, Grade, Fruit Count, Weight, and Firmness.

Step 5 — Twin State Mutation & Explanation: Prediction outcomes and VLM explanations are recorded in the Digital Twin timeline, instantly updating executive charts and readiness gauges.

Step 6 — Harvest Ground Truth Validation: Upon picking, actual refractometer Brix and harvested fruit count are logged in 'Harvest & Evaluation' to benchmark and evaluate model performance.